



Crop Health Monitoring Using AI

Smart Farming for a Sustainable Future

DHANYA S · AMB2062

Imagine detecting plant disease before it's visible to the eye — that farmer now exists, powered by Artificial Intelligence.

The Problem With Traditional Farming

Late Detection

Diseases identified only after visible damage

Inefficient Resources

Over-use of water, fertilisers & pesticides

Manual Monitoring

Labour-intensive, prone to human error

Unpredictability

Environmental conditions remain difficult to forecast

-  Traditional farming treats illness after symptoms appear – AI diagnoses crops **before they feel sick.**



Research Methodology

Quantitative

- Satellite imagery datasets
- IoT sensor field data
- Detection accuracy benchmarks

Qualitative

- Farmer feedback & interviews
- Case studies in AI-based farming
- Traditional vs. AI monitoring comparison

We trained AI like an experienced farmer – learning from thousands of fields to recognise patterns invisible to the human eye.

Data: The Crop's Vital Signs



Satellite & Drone Imagery

Multispectral visual data at scale

Soil Sensors

Moisture, nutrients, pH in real time

Weather Streams

Temperature, rainfall, humidity feeds

Growth Stage Tracking

Multi-source, continuous, real-time updates

📄 Think of the farm as a patient — the data are its **vital signs**.

The AI Framework



Core AI Techniques

- **Computer Vision** – disease detection from imagery
- **Machine Learning** – yield prediction models
- **Predictive Analytics** – environmental risk scoring

- ❏ AI becomes the farmer's digital assistant – watching every leaf, analysing every change, and prescribing the right action at the right time.



Results: What AI Delivers

95%

**Detection
Accuracy**

Early disease
identification before
visible symptoms

30%

**Pesticide
Reduction**

Precision targeting
replaces blanket
application

20%

Yield Increase

Optimised resource use
boosts output and
quality

AI detects leaf discoloration before visible damage, predicts drought stress using soil + weather data, and recommends precise irrigation schedules.

Conclusion

From Reactive → Proactive

AI shifts farming from damage control to intelligent prevention

Precision Agriculture

Right input, right place, right time — every time

Sustainable Impact

Lower environmental footprint, higher productivity



Higher Productivity

Better yields with fewer inputs



Reduced Impact

Less chemical runoff, healthier land



Smarter Decisions

Data-driven farmer empowerment

With AI, farming is no longer just about hard work — it's about **smart, informed decisions**.

Questions for Discussion

1

Can AI replace traditional farming knowledge?

Or does it work best as a complement to lived experience?

2

What barriers exist for AI adoption?

Infrastructure, cost, literacy – where do we start?

3

How can small-scale farmers benefit?

Low-cost sensors, mobile tools, and cooperative data models

Let's discuss how technology can grow the future of agriculture.



Thank You

Presented by Dhanya S · AMB2062

The future of farming is not just green — **it's intelligent.**

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SMART FARMING · SUSTAINABLE FUTURE